

ABHINAVA SAI TIRUNAGARI

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SUMMARY

- AI Engineer with nearly 3 years of experience building Python-based AI systems, RESTful APIs, and microservices using FastAPI, Java, JavaScript, and Git-backed workflows. Delivered production RAG pipelines with embedding models, vector databases, and unstructured data processing for NLP and chatbot-style agent use cases, which improved clinical prediction accuracy from ~40% to 76% (average across multiple evaluation metrics). Strong in prompt engineering, unit testing, and debugging model behavior versus code defects so releases stay reliable under real user traffic. Comfortable translating domain requirements into tested features with attention to edge cases, data confidentiality, and output quality.

TECHNICAL SKILLS

AI Engineering: Python, Machine Learning, NLP, Chatbots, LLM APIs (OpenAI/Claude/Llama), LLM Agents, Prompt Engineering, RAG, LangChain, HuggingFace Transformers, PyTorch

API & Software Engineering: RESTful APIs, Microservices, FastAPI, Node.js, Java, JavaScript, TypeScript, Spring Boot, JSON, XML, Git, Unit Testing, TDD, Debugging, Data Structures & Algorithms

Data & Retrieval Systems: Embedding Models, Vector Databases, Neo4j, PostgreSQL, SQL Server, Redis, SQLite, Pandas, NumPy, ETL Pipelines, Unstructured Data Extraction, Event-Driven Architecture

Cloud & Delivery: AWS, Docker, Kubernetes, GitHub Actions CI/CD, Agile, Go, React.js, Next.js, HTML/CSS, GraphQL

PROFESSIONAL EXPERIENCE

Student Assistant, Quality & Patient Safety | University of Florida
Gainesville, FL

Jan 2026 – June 2026

- Built Python and FastAPI AI services using LLM agents and RAG over unstructured clinical records with embedding models and a Neo4j knowledge graph, raising the causal-relationship percentage captured in the knowledge graph from 9.9% to 40.9% and improving prediction accuracy from ~40% to 76% (an average across multiple evaluation metrics) for safer triage decisions.
- Developed prompt engineering workflows with few-shot examples and strict output formatting, which reduced malformed model responses and made downstream JSON parsing stable for analyst review tools.
- Diagnosed model behavior versus backend defects by tracing retrieval context, prompt templates, and API logs, allowing faster troubleshooting when responses failed due to code paths rather than model quality.
- Implemented unit testing for extraction utilities and API handlers, including edge cases around null fields and noisy text so releases handled clinical data variability without breaking.

Full Stack AI Intern | Cognera Health
Remote, USA

July 2025 – May 2026

- Engineered chatbot-style agent features using Python, FastAPI, and React with JavaScript and TypeScript, enabling care teams to query unstructured documents through audited NLP workflows.
- Created RESTful APIs and microservices that consumed and produced JSON payloads for document ingestion and review, which simplified integration with partner systems and reduced manual handoffs.
- Learned care-review terminology and prior-authorization document workflows from clinical stakeholders, then tailored retrieval prompts and citation rules so nurses could verify agent answers during case review.
- Partnered with clinical analysts and backend engineers in two-week sprints to scope RAG stories, communicate retrieval constraints and data quality needs, and ship tested increments on release cadence.
- Set up Git version control standards with GitHub Actions CI/CD and unit testing gates, improving release confidence by catching schema and parsing regressions before deployment.
- Processed unstructured data with RAG pipelines backed by vector search and embeddings, giving users grounded answers with source snippets instead of unsupported free-form responses.
- Added safeguards for confidentiality and questionable prompt handling by masking sensitive fields and flagging risky requests, supporting ethical AI usage in healthcare contexts.

Member Technical | ADP
Hyderabad, India

July 2023 – June 2024

- Delivered Java and Python RESTful APIs for payroll and finance workflows serving 10,000+ users, using PostgreSQL and SQL Server to keep transaction processing consistent and auditable.
- Built event-driven services with Redis and structured JSON data contracts so dependent teams could integrate faster and recover cleanly from partial failures.
- Implemented XML payroll-import validation adapters that transformed partner XML payloads into internal JSON service contracts, preventing malformed submissions from reaching payroll transaction processors.
- Strengthened unit and integration testing across service modules, improving defect detection for edge conditions in deductions, reimbursements, and multi-step approval logic.
- Resolved production incidents by separating algorithmic issues from implementation bugs during root-cause analysis, which shortened fix cycles for high-priority payroll runs.

Research Intern | ADRIN-ISRO
Hyderabad, India

Jan 2023 – May 2023

- Trained PyTorch deep learning models for satellite image classification and achieved 85% accuracy with 90% precision, enabling faster screening of geospatial imagery for analysts.

- Wrote Python data pipelines to clean and structure large image batches, reducing preprocessing inconsistencies that previously affected model behavior during experiments.
- Applied debugging techniques to isolate data leakage and class imbalance effects, then validated fixes with Pytest-based checks to improve experiment reliability.

Intern | Integrhythm (INRY)
Hyderabad, India

June 2022 – Aug 2022

- Automated Python and Java data integration jobs that cut manual processing effort by 50%, allowing teams to focus on exception handling instead of repetitive transforms.
- Standardized ingestion and validation steps with structured rules for lists and dictionary-like mappings, improving data quality before downstream reporting.

PROJECTS

LLM Agent Platform for Unstructured Data Extraction | Python, FastAPI, RAG, Embedding Models, Vector Databases, React, TypeScript, RESTful APIs, AWS, Docker, GitHub Actions Aug 2025

- Built an AI agent platform using RAG, embedding-driven retrieval, and vector search over unstructured documents so users could ask chatbot-style questions with grounded responses.
- Implemented RESTful APIs and microservices for ingestion, extraction, and review with JSON contracts, which improved interoperability with existing analytics services.
- Added prompt engineering templates with few-shot examples and response formatting rules to keep outputs deterministic for downstream validation logic.
- Used dictionary caches for document metadata and list-based retrieval queues to avoid duplicate chunk fetches in multi-hop flows, reducing agent response latency during repeated follow-up questions.

GraphRAG Biomedical QA System | Python, FastAPI, LangChain, LanceDB, FAISS, Neo4j, HuggingFace Transformers, NLP May 2025

- Designed a multi-stage NLP pipeline combining graph traversal and vector database retrieval to answer biomedical questions from unstructured sources with provenance.
- Compared retrieval failures against prompt and model behavior traces to distinguish ranking defects from generation defects, so engineers fixed the responsible component directly instead of repeatedly tuning the wrong layer.
- Exposed the system through FastAPI endpoints and structured JSON outputs so external clients could integrate without custom parsers.

AI Commerce Insights Platform | Python, FastAPI, React, TypeScript, PostgreSQL, Neo4j, LangChain, RAG, AWS, Docker June 2025

- Built microservices that transformed PDFs, CSV files, and text notes into searchable embeddings and graph entities, enabling tailored agent responses for operations teams.
- Added asynchronous ingestion with quality checks for malformed records and missing fields, improving data handling reliability under mixed document formats.

Event-Driven Data Workflow Engine | Java, Python, FastAPI, Redis, PostgreSQL, Docker, GitHub Actions, Unit Testing Apr 2025

- Created an event-driven microservices workflow using Java and Python for ingestion, transformation, and loading, reducing delayed batch retries through stateful orchestration.
- Enforced unit testing around parser logic and retry behavior, covering edge cases that could duplicate records during transient failures.

Full-Stack Financial Audit & Data Pipeline Platform | React, TypeScript, JavaScript, Python, FastAPI, RESTful APIs, PostgreSQL, SQL Server, Redis Mar 2025

- Delivered a full-stack audit platform with RESTful APIs and JSON-based integrations for transaction reconciliation and anomaly review workflows.
- Integrated AI-assisted checks for unstructured notes and exception descriptions, helping analysts detect suspicious patterns with less manual reading.

Delphi LLVM Compiler & Interpreter Toolkit | Java, JavaScript, ANTLR4, LLVM IR, WebAssembly, Data Structures & Algorithms Feb 2025

- Built a Java compiler and interpreter pipeline that applied trees and graph-style traversals for parsing and code generation, reinforcing strong data structures and algorithms practice.
- Implemented deterministic debugging utilities for token streams and scope resolution, making it easier to separate grammar errors from runtime logic issues.

EDUCATION

University of Florida Aug 2024 – May 2026
Master of Science in Computer Science | GPA: 3.81/4.0 Gainesville, FL

Mahindra Ecole Centrale Aug 2019 – May 2023
Bachelor of Technology in Computer Science Engineering | GPA: 8.94/10.0 Hyderabad, India

CERTIFICATIONS

- Claude 101
- Claude AI Fluency
- AWS Certified Cloud Practitioner
- Microsoft Certified: Azure Fundamentals (AZ-900)